

Root-Locus Lead Compensation

Adding phase lead to pull the locus through a faster pole location.

Ogata, *Modern Control Engineering*, Ch. 6.

In this tutorial you will:

- find the angle deficiency at a desired dominant pole s_d ,
- place the compensator zero and pole to supply that angle, and
- set the gain by the magnitude condition and verify the design.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A phase-lead compensator

$$G_c(s) = K_c \frac{s+z_c}{s+p_c}, \quad |z_c| < |p_c|$$

adds positive phase near the desired closed-loop pole location, "pulling" the root locus to the left and improving transient response (faster rise/settling time, reduced overshoot) at the cost of some bandwidth/noise-sensitivity increase.

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Uncompensated Plant

$G(s) = \frac{4}{s(s+2)}$, with a design specification of $\zeta = 0.5$ (about 16% overshoot) and a faster settling time than the uncompensated system provides.

```
G = tf(4,[1 2 0]);

zeta = 0.5;
% Desired dominant closed-loop poles for a given settling time spec,
% e.g. ts = 1 s (4/(zeta*wn) = 1 => wn = 8)
wn_desired = 8;
sd = -zeta*wn_desired + 1j*wn_desired*sqrt(1-zeta^2);
fprintf('Desired dominant pole: %.4f + %.4fj\n', real(sd), imag(sd))
```

```
Desired dominant pole: -4.0000 + 6.9282j
```

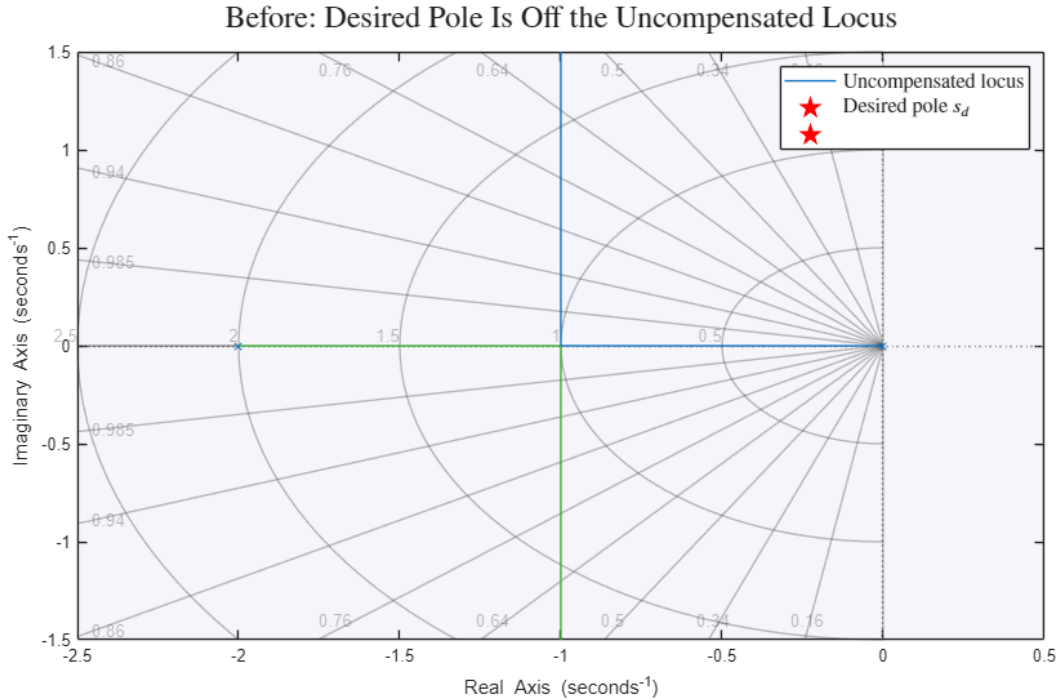
Before: The Uncompensated Locus Misses the Target

Plot the uncompensated root locus and mark the desired dominant pole s_d (both conjugates). It does not lie on the locus, so no value of plain gain K can place a closed-loop pole there – which is exactly why a compensator is needed.

```

figure
rlocus(G)
hold on
plot(real(sd),imag(sd),'rp','MarkerSize',14,'MarkerFaceColor','r')
plot(real(sd),-imag(sd),'rp','MarkerSize',14,'MarkerFaceColor','r')
hold off
legend('Uncompensated locus','Desired pole $s_d$', 'Interpreter','latex','FontSize',12)
title('Before: Desired Pole Is Off the Uncompensated Locus','Interpreter','latex','FontSize',18)
grid on

```



Angle Deficiency

The open-loop transfer function has no finite zeros and a positive DC gain, so its phase at s_d is $\angle G(s_d) = -\sum_i \angle(s_d - p_i)$. For s_d to lie on the **compensated** locus, the lead network must add just enough phase that the total is -180° :

$$\phi_{lead} = -180^\circ - \angle G(s_d).$$

```

[~,p] = tfdata(G,'v');
ol_poles = roots(p);
phase_G = -sum(angle(sd - ol_poles))*180/pi; % angle of G(sd), degrees
phi_lead = mod((-180 - phase_G) + 360, 360); % lead phase to add (deg)
fprintf('Open-loop phase at s_d = %.2f deg\n', phase_G)
fprintf('Phase the lead compensator must add = %.2f deg\n', phi_lead)

```

```

Open-loop phase at s_d = -226.10 deg
Phase the lead compensator must add = 46.10 deg

```

Placing the Compensator Zero and Pole

A common design choice places the compensator zero directly beneath the desired pole's real part (zero at $s = \text{Re}(s_d)$), then solves for the pole that supplies the remaining angle. With the zero at $-z_c$ the zero contributes $\angle(s_d + z_c)$, so the pole must contribute $\angle(s_d + p_c) = \angle(s_d + z_c) - \phi_{lead}$.

```

zc = -real(sd); % zero at s = Re(s_d) (left-half plane)
angle_zero = angle(sd + zc)*180/pi;
angle_pole = angle_zero - phi_lead; % required angle from the compensator pole
pc = -real(sd) + imag(sd)/tand(angle_pole); % pole at -pc on the real axis
fprintf('Lead compensator zero at s = %.4f\n', -zc)
fprintf('Lead compensator pole at s = %.4f\n', -pc)

Gc = tf([1 zc],[1 pc]);

Lead compensator zero at s = -4.0000
Lead compensator pole at s = -11.2000

```

Gain Selection via the Magnitude Condition

```

Kc = 1/abs(evalfr(Gc*G,sd));
fprintf('Compensator gain Kc = %.4f\n', Kc)

```

```

Compensator gain Kc = 20.8000

```

Verification

Confirm the design: with the chosen K_c , a closed-loop pole should sit essentially on top of the desired location s_d .

```

T_check = feedback(Kc*Gc*G,1);
cl_poles = pole(T_check);
[~,idx] = min(abs(cl_poles - sd));
fprintf('Desired s_d = %.4f + %.4fj; nearest closed-loop pole = %.4f + %.4fj\n', ...
    real(sd), imag(sd), real(cl_poles(idx)), imag(cl_poles(idx)))

Gc_full = Kc*Gc;
G_comp = series(Gc_full,G);
T_comp = feedback(G_comp,1);
T_uncomp = feedback(G,1);

```

```

Desired s_d = -4.0000 + 6.9282j; nearest closed-loop pole = -4.0000 + 6.9282j

```

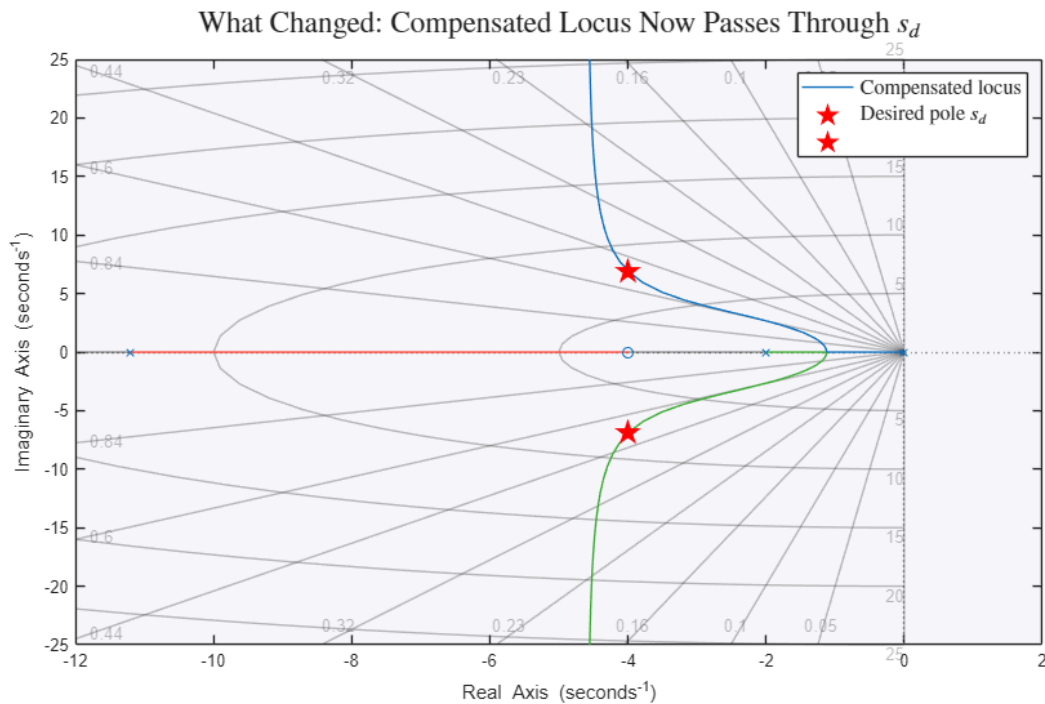
What I Did: The Compensator Reshapes the Locus

Adding the lead pole/zero pulls the locus to the left so that it now passes through the desired pole s_d (marked again for reference).

```

figure
rlocus(G_comp)
hold on
plot(real(sd),imag(sd),'rp','MarkerSize',14,'MarkerFaceColor','r')
plot(real(sd),-imag(sd),'rp','MarkerSize',14,'MarkerFaceColor','r')
hold off
legend('Compensated locus','Desired pole $s_d$','Interpreter','latex','FontSize',12)
title('What Changed: Compensated Locus Now Passes Through $s_d$', 'Interpreter','latex','FontSize',17)
grid on

```



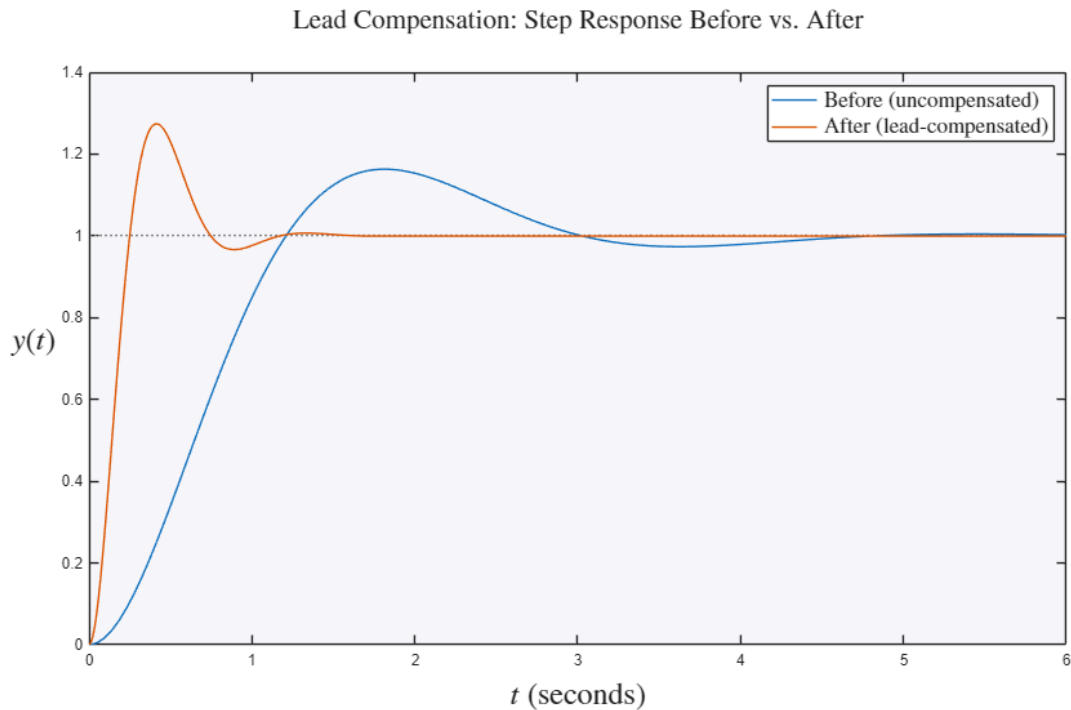
After: What Changed in the Time Response

Side by side, the lead compensator speeds up the response and cuts the overshoot relative to the uncompensated closed loop.

```
figure
hold on
step(T_uncomp)
step(T_comp)
hold off
legend('Before (uncompensated)', 'After (lead-compensated)', 'Interpreter', 'latex', 'FontSize', 14)
title('Lead Compensation: Step Response Before vs. After', 'Interpreter', 'latex', 'FontSize', 18)
ylabel('$y(t)$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)

info_before = stepinfo(T_uncomp);
info_after = stepinfo(T_comp);
fprintf('Overshoot:    before = %.1f%%, after = %.1f%%\n', info_before.Overshoot, info_after.Overshoot)
fprintf('Settling time: before = %.3f s, after = %.3f s\n', info_before.SettlingTime, info_after.SettlingTime)

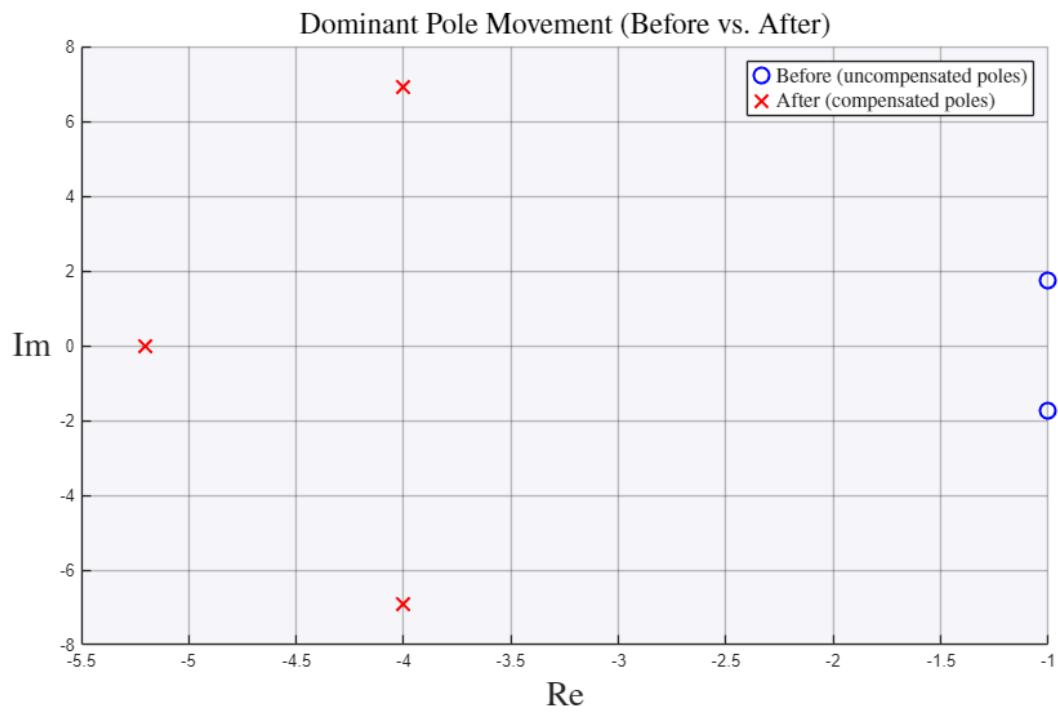
Overshoot:    before = 16.3%, after = 27.4%
Settling time: before = 4.038 s, after = 1.025 s
```



After: Where the Dominant Poles Moved

The pole map makes the design goal explicit – the dominant closed-loop poles move from the slow uncompensated location to the faster, better damped target near s.d.

```
figure
hold on
plot(real(pole(T_uncomp)),imag(pole(T_uncomp)),'bo','MarkerSize',9,'LineWidth',1.5)
plot(real(pole(T_comp)),imag(pole(T_comp)),'rx','MarkerSize',11,'LineWidth',1.5)
hold off
grid on
legend('Before (uncompensated poles)','After (compensated poles)','Interpreter','latex','FontSize',12)
title('Dominant Pole Movement (Before vs. After)','Interpreter','latex','FontSize',17)
ylabel('$\mathrm{Im}$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$\mathrm{Re}$','Interpreter','latex','FontSize',20)
```



Try it yourself

- Ask for a faster pole ($\omega_{n_desired} = 12$) and notice the compensator pole move further left and the gain K_c grow.
- Place the zero away from $\text{Re}(s_d)$ and see the design still succeed but need a different pole angle.